

**OFFLINE FIRST DISEASE DETECTION AND
CLASSIFICATION IN PUMPKIN LEAF USING
ENSEMBLE MODEL**

BY

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20234419568

**A THESIS SUBMITTED TO THE POSTGRADUATE
SCHOOL FEDERAL UNIVERSITY OF TECHNOLOGY,
OWERRI**

**IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE AWARD OF THE DEGREE (MASTER OF
SCIENCE), M.Sc. IN COMPUTER SCIENCE**

JUNE, 2025

CERTIFICATION

This is to certify that this work: Offline-first Disease Detection and Classification in Pumpkin Leaf using Ensemble Model was carried out by Njoku, Okechukwu, Reg. number (20234419568) in partial fulfilment for the award of the degree of M.Sc. in Computer Science in the Department of Computer Science of the Federal University of Technology, Owerri.

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APPROVAL

This project entitled DISEASE DETECTION AND CLASSIFICATION IN UGU LEAF USING ML / AI carried out by Njoku, Okechukwu Valentine with Registration Number INSERT YOUR REG NO and SURNAME, FIRSTNAME OTHERNAME with Registration Number 20234419568 has been presented to the Department of Computer Science, School of Information and Communication Technology, in partial fulfillment for the award of Master of Science (MSc.) in Computer Science. The undersigned have reviewed this project report and hereby approve it for its contribution to knowledge and adherence to the standard of academics of FEDERAL UNIVERSITY OF TECHNOLOGY OWERRI.

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DEDICATION

This work is dedicated to lovers of effort, dedication, hard-work and the African reawakening.

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my maker for the gift of life. Special thanks goes to Dr. Ayogu Ikechukwu my Postgraduate coordinator, my mentor and boss, without his guide and advice, I would not have started this program.

I would love to appreciate Dr. Okpalla Chidimma, my supervisor, ma, you have been most tolerant and supportive in this journey and I am more grateful than words can explain. My heartfelt appreciation also goes to Dr. O. A. Njoku for her insights and contribution.

I sincerely appreciate all my lecturers, you have shown me knowledge and skills transfer like never seen elsewhere. Your tutelage is second to none.

My unspeakable gratitude goes to Barr. Chima Anozie (aka Homebase), Eng. Enyeribe Emmanuel Mmegwa, Dr. Dorathy Anyanwu and Dr. Francis Ahaotu, and last but not the least, Mrs. Njoku Onyinyechi, my most precious wife for her support during this program.

I want to sincerely appreciate my colleagues - Salvation, Chioma, Damian and Hannah. Hm, I cannot thank all I should, but am most grateful thus far.

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CHAPTER ONE

INTRODUCTION

1.1 Background Information

Pumpkin leaves (*Cucurbita spp*), one of the underutilized crops which belong to the family, curcubita often consumed as a vegetable, includes various species like *Cucurbita pepo*, *C. moschata*, and *C. maxima*. They are known for their nutrition values, with high levels of iron and vitamins A, B, and E are a secondary crop grown alongside other pumpkins (Mohammed et al., 2021). Originating from regions like Mexico, Central America, and South America, pumpkin leaves are now cultivated in many tropical, subtropical, and temperate climates, particularly in West Africa where there are well consumed and cherished, notably known as “ug-boguru” in Igbo language. Pumpkin leaves are rich in iron content which is vital to the proper function of haemoglobin, which reduces the chances of anaemia. Pumpkin leaves also have the ability to reduce incidences of non-communicable chronic disease such as type 2 diabetes. However, pumpkin is easily attacked by fungal (powdery mildew), bacterial (leaf spot), and viral disease (mosaic virus) affecting crop yield and productivity (Yadav et al., 2022). These diseases are traditionally discovered through visual observation, which is time consuming and susceptible to error, especially in early infection stages. Crop disease detection refers to the assessment of disease appearance, intensity, and effects (Abdu et al., 2020). Crop disease can be classified as a dynamic situation that disorganize the proper functioning of plants and hinders their growth during a specific time frame (Abdu et al., 2020). The identification of crop illness involves the use of machine vision techniques to acquire images, which are subsequently subjected to analysis in order to determine the presence or absence of pathogens within the crop (Liu and Wang, 2021). Barure et al. (2020) explained crop disease classification as the techniques of grouping each subject under study into various groups. Due to advancement in technology, Artificial Intelligence (AI) and Convolutional Neural Networks (CNNs) a sub-field in Machine Learning (ML), are enabled to achieve high accuracy in detecting plant and diseases. The use of deep CNNs for real-time detection shows great performance in agricultural environment (Khalid et al., 2023). Nwaneto and Yinka-Banjo (2024) achieved an accuracy of 94% in identifying various crop diseases by comparing MobileNet and custom CNNs on a large dataset. Based on these findings, the potential for integrating AI-driven techniques to pumpkin leaf disease detection provides robust, scalable and cost-effective tools for supporting smallholder farmers.

1.2 Problem Statement

Pumpkin leaf cultivation is affected by fungal (powdery mildew), bacterial (leaf spot), and viral disease (mosaic virus) resulting to low yield of productivity. Timely and accurate identification of these diseases is necessary for proper management and control. Most farmers rely on the traditional manual method using visual observation, which is time consuming, susceptible to error, and depend on expertise knowledge which may not be readily available especially

in early infection stages which pose a gap. In addition, early symptoms of pumpkin appear to be similar at initial stage and may be hard to identify and differentiate with ordinary eyes. Due to the recent technological advancement in AI and ML, an intelligent diagnostic system will be developed for automated disease discovery of the pumpkin leaf and classification especially in rural areas. The integration of AI in disease detection and classification of the pumpkin leaf is necessary to enable accurate, timely and real-time detection, thereby supporting decision making and leading to crop yield.

1.3 Objectives

The aim of the of the research is to design an Offline-first Disease Detection and Classification in Pumpkin Leaf using Ensemble Model to detect and classify pumpkin leaf diseases and suggest remedies using lightweight CNN and Vision Transformer or Attention Mechanism.

Specific Objectives

1. To train a deep learning model to detect diseases in pumpkin leaves using an ensemble of a lightweight CNN and Vision Transformer (ViTt).
2. To validate the model accuracy
3. To deploy the model to cloud and or mobile for public access and use.

Application Programming Interface (API) for integration in other applications and devices. The objectives of the study is primarily to develop an accurate, efficient and robust system for identifying the diseases mentioned in 1.5 particularly in its early stage. This early detection is very important for prompt management intervention and control and can assist in reduction of crop losses and prevent the spread of the disease(s).

1.4 Justification of the Study

Early identification of the pathogen type of plant infection is of high significance for disease control. Various methods are used to diagnose pathogens of disease on plant. Traditional methods for diagnosis of plant pathogens rely on visual observation, microscopy, mycological analysis, and biological diagnostics or the use of indicator plants, which are not accurate and lack early detection. Rapid and reliable detection of plant disease and identification of its pathogen is a vital stage in disease control. Early identification of the cause of the disease allows timely selection of the proper protection method and ensures prevention of crop losses. There are a number of traditional methods for identifying plant diseases, however, in order to ensure the promptness and reliability of diagnosis, as well as to eliminate the shortcomings inherent in traditional diagnosis, in recent years, new means and technologies for identifying pathogens have been developed and introduced into practice. This project provides information on such innovative methods of diagnosis of diseases and classification of their pathogens, which are used widely in the world today, such as immunodiagnosics, molecular-genetic (and phylogenetic) identification, mass spectrometry (Khakimov et al., 2022)

1.5 Scope of the Study

This study covers the application of AI/ML methods for the detection, classification and recommendation of treatment options for the main diseases affecting pumpkin (*curcubita spp*) plant only. The diseases covered in this project is downy mildew, powdery mildew, mosaic virus. Also, covered is the deployment on mobile framework or platform via a mobile application.

1.6 Limitations of Existing System

1. Complexity may result to challenge in low-power devices.
2. May lack reliability for other crops apart from the present ones.
3. Classification performance requires improvement
4. Complexity in feature integration may stop model interpretability and adaptability.
5. The dense architectural structure may increase memory consumption

1.7 Research Questions

1. What is the accuracy of using image processing to identify and classify different pumpkin leaf diseases?
2. How does image processing and classification compare to traditional methods of pumpkin leaf disease detection?
3. How can pumpkin leaf disease detection and classification model be deployed on mobile?

CHAPTER TWO

LITERATURE REVIEW

2.1 Conceptual Framework

2.1.1 Concept of Plant Disease Detection

Farmers face major challenges in dealing with crop disease, making classification and examination of illnesses key to optimizing food yield in agriculture (Chen et al., 2021). The study of detecting and identifying plant disease is important, especially as it can potentially monitor extensive crop fields and rapidly identify disease symptoms on plant leaves. Therefore, the desire for a quick, accurate and cost effective approach to predicting instances of crop disease is of utmost importance (Chen et al., 2021). Artificial Intelligence (AI) greatly contributes to the agricultural sector, improving a nation's gross domestic product (GDP). The application of edge intelligence to agriculture, especially through the use of deep learning models like the YOLOv3 neural network on embedded systems such as the NVIDIA Jetson TX2, presents a novel approach. Hyperspectral and multispectral knowledge is widely employed in different application areas in agriculture, including sustainable agriculture and these technologies provides necessary knowledge for farmers and agricultural management (Ang and Seng, 2021)

2.1.2 Traditional Methods of Disease Detection

Traditional plant disease detection is through manual observation by farmers and other agricultural extension workers. However, in majority of the cases, the diagnosis is based on guess on the disease which is performed visually by humans. Trained raters may be efficient in recognizing and quantifying diseases but they have associated limitations that may cause harm to crop yield. Challenges of this method are listed below:

1. Accuracy depends on the experience and accuracy of the observer.
2. Diseases are diagnosed only when symptoms are visible, preventing early intervention.
3. It is difficult to differentiate between diseases with similar symptoms by mere visual observation.
4. Manual inspection is susceptible to error.

2.1.3 Overview of Artificial Intelligence and Machine Learning in Agriculture

Deep Learning in Agriculture

Disease prediction at an early stage is vital to preventing massive crop loss and ensuring greater crop yield. Advancement in technology in terms of speed and power, along with improved access to massive datasets, contribute to the enhanced robustness of disease detection systems (Wu et al., 2020). A hybrid integration of classifier methods showed a recognition rate of 91.11 %, surpassing the performance of serial, parallel, and deep learning approaches (El Massi et al., 2021). In addition, the application of the CovNet algorithm for recognizing

weeds in crops and the use of hybrid deep-learning models demonstrate potential results in terms of accuracy and parameter reduction (Zhou et al., 2021). Deep transfer learning, leveraging pre-trained datasets like Inception and ImageNet modules, proves effective in identifying plant diseases, with the aid of Support Vector Machine (SVM) and multi-layer perceptron contributing to higher accuracy. The deployment of Deep Convolutional Neural Networks for identifying corn dietary sickness and the use of hardware enhancements, such as Raspberry Pi3 with an Intel Movidius Neural Compute Stick, has lead to superior metric accuracy and performance (Sun et al., 2020).

As AI and DL technology advances, Computer Vision (CV) has made significant strides. CV-based approaches employ Principle Component Analysis (PCA) that has proven useful in diagnosing grape leaf diseases with quality research accuracy (Xie et al., 2020). Validation accuracy leveraging VGGNet on real-world datasets reached 91.83% (Xie et al., 2020). Vision Transformer (ViT) has shown potential in developing models for detecting plant diseases. Researchers have examined ViT to identify plant disease on mobile devices and classify plant automatically in real-time. ViT based models out performed prior existing model in classifying benchmark datasets such as CIFAR-100, Oxford-IIIT pets, Oxford Flowers-102, and ImageNet.

Image Based Techniques

Image - based disease detection has become powerful and reliable in modern precision agriculture. This method utilizes digital images of plant leaves and advanced computational algorithms to identify diseases based on features such as colour, textual and shape. With the integration of machine learning and deep learning approach, particularly the CNNs, image based systems can detect and classify disease with high precision. The existing system supports two popular machine learning models, Support Vector Machine (SVM) and Deep Learning. Based on the result suggestions were preferred for image processing based plant leaf disease detection within the constraints of architectural complexity, computational power, and training data. Stages are shown in figure 2.1 below:

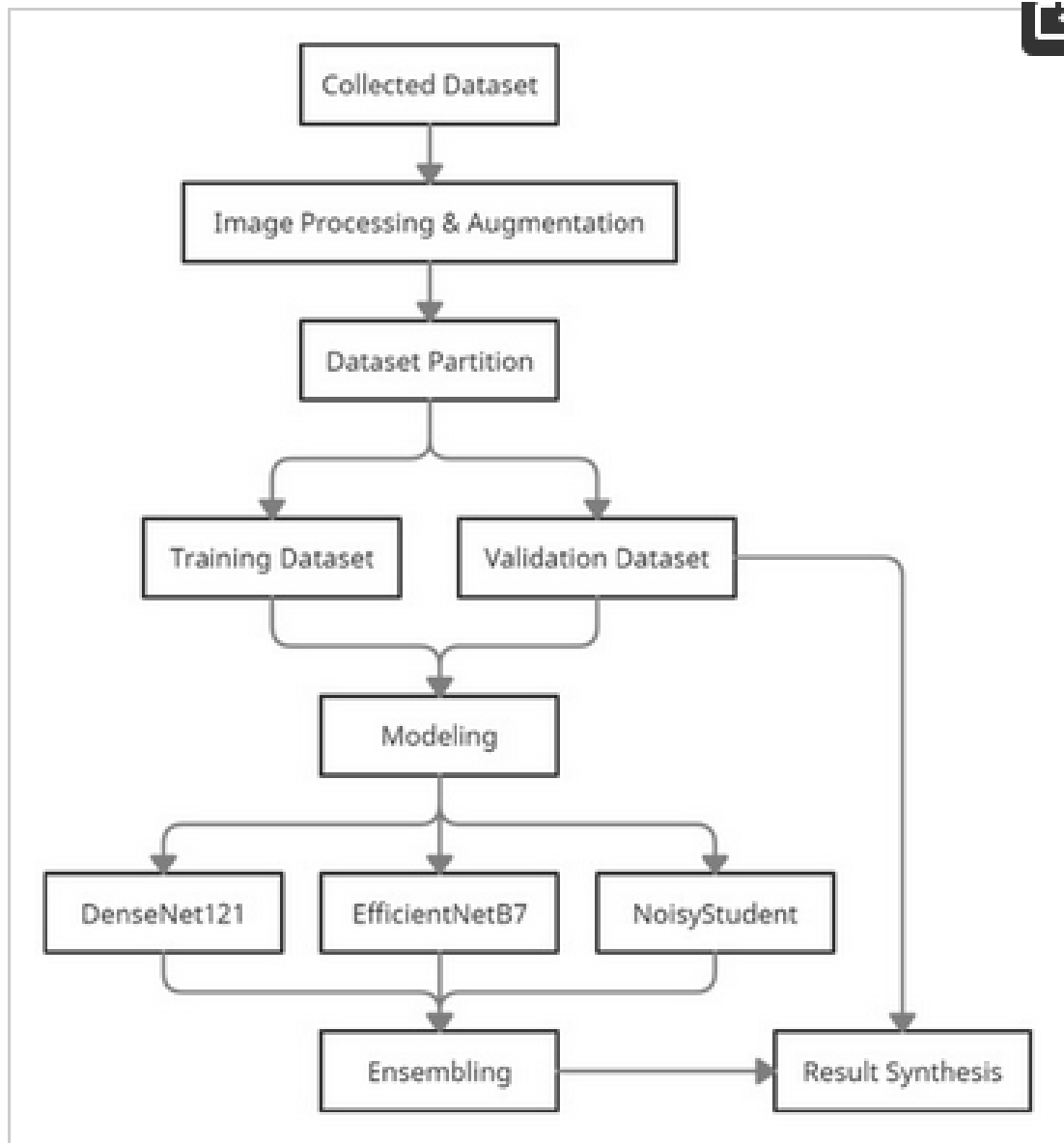


Figure 2.1: Image Based Techniques Steps (Bansal et al., 2021)

Studies on Pumpkin Leaf Disease Detection

The survey of leaf spot disease of fluted pumpkin was carried out in 2019 during farming season. Leaf samples were collected from individual farmers at Ten Telfairia locations where Telfairia were cultivated namely: Apir, Federal Housing Estate, Kanshio, Ichuwa, Welfare quarters, New GRA, Yarkyo, and Lower Benue River Basin, along River Benue Bank and at NASME (Nigeria Army School of Military Engineering) barracks. Disease incidence was examined by expressing the numbers of infected Telfairia plants as a percentage of the total number of Telfairia plants. (Ekhuemelo and Otor, 2021).

2.2 Theoretical Framework

The foundation of this study is built on the theories below:

2.2.1 Decision Theory in Artificial Intelligence

Decision theory is a foundational concept in Artificial Intelligence (AI), which enables machines to make analytical and informed decisions based on available data. It gathers principles from various fields such as mathematics, statistics, economic and psychology to model and improve decision making processes. Decision theory provides a platform to predict outcomes, evaluate choices and guide actions in uncertain environment. Applying this theory in the study will enable AI and ML model to imitate expert diagnosis in recognizing pattern and proffering treatment based on disease type detected.

2.2.2 Pattern Recognition Theory

Pattern Recognition is the process of using machine learning algorithms to recognize patterns. It means sorting data into categories by analyzing the pattern present in the data. One major benefit of pattern recognition is that it can be used in many areas. In pattern recognition application, raw data is processed and converted to the form the machine can use. The two major method involved in pattern recognition is clustering and classification. Pattern recognition will allow for extraction of disease feature in pumpkin leaf images and classify diseases based on learned visual variances.

2.3 Empirical Framework

2.3.1 Related Work

Plant disease detection through machine learning and deep learning models have been an interesting area of recent research. Jiang et al. (2019) developed a deep CNN architecture for identifying apple leaf disease. The architecture integrates CNN with the inception module to recognize five types of apple leaf disease. Bukumira et al. (2022) proposed a carrot grading system using computer vision techniques and a CNN graph. These identified frameworks showed the capability to classify carrot fruit accurately and grade them. In addition, other studies have suggested DL models for recognizing apple leaf diseases including Ful et al. (2022), Bansal et al. (2021) and Khan et al. (2022) To enhance accuracy, the models employ different methods like image augmentation, attention mechanisms, and dilated convolution. Studies have examined hybrid deep-learning models in diverse fields, such as agriculture and medical sector, for disease diagnosis and classification (Zhang et al., 2023) and (Waleed et al., 2022). These investigations have proved the efficacy of combining deep learning architectures to accurately extract relevant features from the input data. Hosny et al. (2023) designed a hybrid approach using lightweight deep model features and Local Binary Pattern features to detect and classify leaf disease in grapes, apples and tomatoes. These techniques resulted in accuracy rate of 97.8%, 97% and 96% for each types of plants respectively. (Ahila et al., 2019) create CNN model by adjusting the LeNet model to classify corn leaf disease. A corn leaf image dataset from PlantVillage was utilized to train their method. The techniques resulted in an accuracy of 97.89%. (Waheed et al., 2020) developed an optimized DenseNet structure to detect and classify disease which includes northern leaf blight, common rust, gray leaf spot, and healthy leaves leading to an accuracy of 98.06%. Wu et al. (2019) proposed a CNN model to detect northern leaf blight on corn leaves using UAV images. They used the pre-trained ResNet architecture for feature extraction, employing its result as inputs to a basic linear classifier leading to an overall accuracy of 95.1.

2.4 Summary of Related Works for Plant Leaf Disease Detection and Classification

Table 2.1: Summary of Related Works for Plant Leaf Disease Detection and Classification

References	Objective	Method	Key Finding	Limitations
Hosny et al. (2023)	Multi-class plant disease classification	Hybrid of CNN and Local Binary Pattern	Improved multi-class classification accuracy	Complexity in feature integration may stop model interpretability and adaptability.
Bukumira et al. (2022)	Graph Convolutional Neural Network (GCNN) and computer vision on carrot grading	Cascaded graph CNN and Bayesian Optimisation	Automated grading of carrot	Limited carrot grading only. The GCNN may be computationally intensive for real-time grading
Fu et al. (2022)	Lightweight model for apple leaf disease detection	Lightweight CNN	Reduced model complexity with increased accuracy	Limited to apple leave, may lack reliability for other crops.
Khan et al. (2022)	Real-time apple detection system	Lightweight-CNN and YOLOv4	Real-time performance	Classification performance requires improvement.
Bansal et al. (2021)	Apple leaf disease detection	Ensemble of pre-trained deep CNN models.	Accuracy and enhanced classification for apple disease.	Multiple diseases may reduce accuracy.

Table 2.1 – continued from previous page

References	Objective	Method	Key Find- ing	Limitations
Waheed et al. (2020)	Disease recognition and classification of corn leaf	Optimized DenseNet Model	Enhanced classification effectiveness and accuracy	The dense architectural structure may increase memory consumption.
Ahila et al. (2019)	Classification of maize leaf diseases	Binary pattern modified LeNet	Real-time performance with Lightweight model	Limited to maize leaf disease classification.
Jiang et al. (2019)	Real-time detection of disease on apple leaf	Enhanced CNN	Provide high quality performance solution with real time detection	Complexity may result to challenge in low-power devices.
Wu et al (2019)	Autonomous detection of plant disease	CNN model on aerial imagery	Robust in large-scale disease detection	Limited to aerial imagery.

2.4.1 Limitations of Existing System

1. Complexity may result to challenge in low powered devices.
2. May lack reliability for other crops apart for the preset ones.
3. Classification performance requires improvement.
4. Complexity in feature integration may limit model easy of interpretation.
5. The dense architectural structure may increase memory consumption.

2.4.2 Research Gaps

The related work shows gap in research investigating only one disease type on a particular leaf plant. The existing system can detect only one type of disease on pumpkin leaf. And this project is to fill this gap by covering four different diseases of the pumpkin leaf.

CHAPTER THREE

METHODOLOGY

3.1 Methodological approach

This study following the experimental and reproducible approach for the image-based disease detection on pumpkin leaves. The workflow is

1. Data collection and labelling - the data was collected from kaggle (make it a link) and it was already labelled.
2. Preprocessing and augmentation - this was transformed to make the dataset on-device friendly
3. Two lightweight models trained independently:
 - a lightweight CNN (mobileNetV2-style)
 - a compact Vision Transformer (simple ViT / MobileViT-like)
4. Ensemble inference by averaging predicted probabilities.
5. Model optimisation and conversion to mobile-friendly format tensorflow-lite (TFLite) with post-training quantization for offline use.
6. Offline-first deployment: a local mobile app (android/iOS/Flutter) or small embedded device like raspberry Pi using Tensorflow lite model and a tiny local API for batch / real - time inference.

The system emphasis is on low memory, low latency and ability to run fully offline on mobile devices and edge compute. The address gaps identified in the literature and the existing system analysis.

3.2 Materials

3.2.1 Hardware

- Development terminal for training (in addition to google colab)
- Target inference devices:
 - Android smartphone (ARM64)
 - Raspberry Pi 4 (for offline kiosk/edge)
 - Low-memory ($\geq$ 1GB RAM recommended)

3.2.2 Software

- Python 3.9+
- Tensorflow 2.10+ or 2.11 (for tensorflow lite conversion)
- Kera (integrated in TF)
- Numpy, pandas, scikit-learn (for metrics and PCA)
- OpenCV / Pillow (image I/O)
- Flask (optional small local server for offline-first desktop/edge)
- Android / Flutter for mobile front-end (integration with TFLite)

3.2.3 Dataset

- Pumpkin leaf images labelled, collected from kaggle (kept locally). The dataset has the following directory structure per class:

```
/
├── train
│   ├── healthy
│   ├── powdery_mildew
│   ├── leaf_spot
│   └── mosaic_virus
├── val
│   ├── healthy
│   ├── powdery_mildew
│   ├── leaf_spot
│   └── mosaic_virus
└── test
    ├── healthy
    ├── powdery_mildew
    ├── leaf_spot
    └── mosaic_virus
```

- Images should be stored locally on the device for offline operations.

3.3 Methods - Overview of processing steps

Image Preprocessing • Resize images to 224x224 (mobile-friendly)

- Image standardization / scaling to range [0,1]
- Basic augmentation (rotation, flip, brightness) done

Feature extraction and model design • **Lightweight CNN: MobileNetV2 backbone (or MobileNetV3 if available) with a small classifier head**

- **Lightweight ViT: Patch embedding + small transformer encoder stack (eg. 4 heads, 6 layers but reduced dims).**

- **Both models produce class probabilities**

Ensemble strategy • Average class probabilities from both models.

- Optionally, weighted average (weights chosen using validation set)

Training • Use Adam optimizer, categorical crossentropy, early stopping and model checkpointing

- Use class weighting or balanced sampling if dataset is imbalanced.

Offline & Mobile Optimization • Exported best model to TFLite (to make model compatible with mobile architecture)

- Apply post-training quantization (dynamic or full integer) to reduce size and latency.
- Validate quantized model accuracy on validation/test sets

Deployment • Provide TFLite to mobile for offline inference. Provide a minimal offline Flash server (for kiosk) that loads the TFLite model and returns results locally.

3.4 Image Preprocessing and Augmentation

- Resize images to 224x224 (for both CNN and ViT)
- On-the-fly augmentation for training:
 - random horizontal/vertical flips
 - random rotation $\pm 20^\circ$
 - random brightness/contrast jitter
 - random crops followed by resize (optional)
- For ViT, we still use 224x224 but patchify (eg 16x16 patches).
- Normalize images using ImageNet mean/std (or scale to [0,1] and use batch normalization).

3.5 Model Architectures

3.5.1 Lightweight CNN (MobileNetV2-based)

- Use `Keras tf.keras.applications.MobileNetV2` without top.
- `GlobalAveragePooling2D` + small dense head (512 -> dropout -> num_classes).
- Use `alpha = 0.35` or `alpha = 0.5` to reduce parameters making it mobile friendly

3.5.2 Lightweight ViT(compact from scratch)

- Patch embedding: Conv2D with kernel/strid = patch_size (eg 16)
- Flatten patches -> Dense to projection_dim
- Add a small class token + positional embeddings.
- Use a stack of TransformerEncoder blocks (multi-head attention, small MLP).
- Keep projection_dim small (eg. 128) and number of heads small (4) to limit compute

3.5.3 Ensemble

- Average softmax probabilities from both models.
- Save the ensemble procedure as part of the inference wrapper – on mobile we run both TFLite modes and average outputs.

3.6 Evaluation Metrics

The metrics used to evaluate the model performance are:

- Accuracy
- Precision
- Recall
- F1-score (per class)
- Confusion matrix
- Model size (in MB) and inference time (ms) on target device. Memory usage and peak CPU utilization (optional)

3.7 Offline-first design choice

• Store the TFLite model(s) bundled inside the mobile app or side-loaded to a local directory so the system functions without network. • Use local device storage for captured images and inference logs. • Provide an optional sync mechanism (manual or background when on Wi-Fi) to push anonymized inference logs or model updates to a central server. • Keep TM (threading) and batching small to avoid blocking UI and to conserve battery.

3.8 Reproducible Code (training, ensemble creation, TFLite conversion) Below is working Python code that you can run locally. It implements: • data generators, • MobileNetV2 model, • a compact ViT model, • ensemble training loop (train separately; combine during inference), • TFLite export with quantization, • a tiny Flask offline inference server example.

```
my_variable = 10 print(f"The value of my_variable is: {my_variable}")
```

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